CS1231S Assignment #2

AY2023/24 Semester 2

Deadline: Monday, 8 April 2024, 1:00pm

IMPORTANT: Please read the instructions below.

This is a graded assignment worth 10% of your final grade. There are **six questions** ("question" 0 and five problem-solving questions) with a total score of 40 marks. Please work on it by yourself, not in a group or in collaboration with anybody. Anyone found committing plagiarism (submitting other's work as your own), or sending your answers to others, or other forms of academic dishonesty will be penalised with a straight zero for the assignment, and possibly an F grade for the module. Please see SoC website "Preventing Plagiarism" https://www.comp.nus.edu.sg/cug/plagiarism/.

You are to submit your assignment to **Canvas > Assignments** before the deadline.

Your answers may be typed or handwritten. Make sure that it is legible (for example, don't use very light pencil or ink if it is handwritten, or font size smaller than 11 if it is typed) or marks may be deducted.

You are to submit **a SINGLE pdf file**, where each page is A4 size. Do not submit files in other formats. If you submit multiple files, only the last submitted file will be graded.

Late submission will NOT be accepted. We will set the closing time of the submission folders to slightly later than 1pm to give you a few minutes of grace, but in your mind, you should treat **1pm** as the deadline. If you think you might be too busy on the day of the deadline, please submit earlier. Also, avoid submitting in the last minute; the system may get sluggish due to overload and you will miss the deadline.

Note the following as well:

- Name your pdf file with your **Student Number**. Your student number begins with 'A' (eg: A0234567X). (Do not mix up your student number with your NUSNET-id which begins with 'e'.)
- At the top of the first page of your submission, write your **Name** and **Tutorial Group**.
- To keep the submitted file short, please submit your answers without including the questions.
- As this is an assignment given well ahead of time, we expect you to work on it early. You should submit **polished work**, not answers that are untidy or appear to have been done in a hurry, for example, with scribbling and cancellation all over the places. Marks may be deducted for untidy work.
- It is always good to be clear and leave no gap so that the marker does not need to make guesswork on your answer. When in doubt, the marker would usually not award the mark.
- Do not use any methods that have not been covered in class. When using a theorem or result that has appeared in class (lectures or tutorials), please quote the theorem number/name, the lecture and slide number, or the tutorial number and question number in that tutorial, failing which marks may be deducted. Remember to use numbering and give justification for important steps in your proof, or marks may be deducted.

To combine all pages into a single pdf document for submission, you may find the following scanning apps helpful if you intend to scan your handwritten answers:

* for Android: https://fossbytes.com/best-android-scanner-apps/

* for iphone:
https://www.switchingtomac.com/tutorials/ios-tutorials/the-best-ios-scanner-apps-to-scan-documents-images/

If you need any clarification about this assignment, please do NOT email us or post on telegram, but post on **QnA "Assignments"** topic so that everybody can read the answers to the queries.

Question 0. (Total: 2 marks)

Check that …

- you have submitted a pdf file with your Student Number as the filename. [1 mark]
- you have written <u>both</u> your name and tutorial group number (eg: T02) at the top of the first page of your file. (If you miss either one you will not be given any mark.) [1 mark]

Question 1. Functions (Total: 7 marks)

Define a function $g : \mathbb{Z} \to \mathbb{Z}$ by setting, for each $x \in \mathbb{Z}$,

$$g(x) = \begin{cases} \dfrac{x}{2}, & \text{if } x \text{ is even;} \\ 5x + 3, & \text{if } x \text{ is odd.} \end{cases}$$

(a) Is g injective? Answer "yes" or "no", followed by a proof or counterexample. [2 marks]

(b) Is g is surjective? Answer "yes" or "no", followed by a proof or counterexample. [2 marks]

Given a function $f(x)$, we define the function $f^{(n)}(x)$ to be the result of n applications of f to x, where $n \in \mathbb{Z}^+$. For example, $f^{(3)}(x) = f\left(f\big(f(x)\big)\right)$.

(c) What is $g^{(6)}(10)$? Show your working. [1 mark]

We also define the *order* of an input x with respect to f to be the smallest positive integer m such that $f^{(m)}(x) = x$.

(d) What is the order of 2 with respect to the function g? Show your working. [2 marks]

Question 2. Functions (Total: 8 marks)

A function $f : \mathbb{R} \to \mathbb{R}$ is said to be *linear* if, and only if,

$$\forall x, y \in \mathbb{R}\; f(x + y) = f(x) + f(y) \quad \text{and} \quad \forall \lambda, x \in \mathbb{R}\; f(\lambda x) = \lambda f(x).$$

Let $f : \mathbb{R} \to \mathbb{R}$ be a linear function.

Define the relation ~ on $\mathbb{R}$ such that $\forall a, b \in \mathbb{R}\; a \sim b \Leftrightarrow a - b \in f^{-1}(\{0\})$.

(a) Prove that ~ is an equivalence relation. [5 marks]

(b) Provide a well-defined surjective function $\pi : \mathbb{R} \to \mathbb{R}/\sim$. (You need to prove that your function is (1) well-defined, and (2) a surjection.) [3 marks]

Question 3. Mathematical Induction (Total: 7 marks)

Let $v_0, v_1, v_2, \cdots$ be a sequence defined as follows:

$$v_0 = 0, \qquad v_1 = a \text{ for some } a \in \mathbb{Z}^+, \text{ and} \qquad v_n = v_{n-1} + v_{n-2} \text{ for } n \geq 2.$$

Let $w_n = v_{n-1}^2 - v_{n-2}v_n$ for $n \geq 2$.

(a) Write the values of w_2, w_3, w_4 and w_5. You do not need to show your working. [2 marks]

(b) Make a conjecture for an *explicit formula* (also known as *closed formula*) for w_n in terms of a and n, for $n \geq 2$. [1 mark]

(c) Prove your conjecture in (b) using mathematical induction. (Follow the structure of mathematical induction proofs we used in class. Please do not skip steps or marks will be deducted.) [4 marks]

Question 4. Cardinality (Total: 9 marks)

Let A, B and C be sets.

(a) Prove or disprove: $(A \setminus B) \cup (A \cap B) = A$.

For each of the following statements, state whether it is TRUE or FALSE. If true, provide a proof. If false, provide a counterexample. You may use the following propositions without proof:

Proposition 1: If A is finite, then $A \cap B$ is finite.
Proposition 2: If A and B are finite, then $A \cup B$ is finite.

(b) If A is countably infinite and B is finite, then $A \setminus B$ is countably infinite.

(c) If A and B are both countably infinite, then $A \cap B$ is countably infinite.

(d) Suppose $A \subseteq B \subseteq C$. If A and C are both countably infinite, then B is countably infinite.

Question 5. Counting and Probability (Total: 7 marks)

(a) How many integer solutions are there for the following equation, where each $x_i \geq 1$?

$$x_1 + x_2 + x_3 = 10.$$

Show your working instead of just writing out the final answer. [1 mark]

(b) The figure on the right shows a combination lock with 40 positions. To open the lock, you rotate to a number in a clockwise direction, then to a second number in the counterclockwise direction, and finally to a third number in the clockwise direction. If consecutive numbers in the combination cannot be the same, how many three-number codes can there be? Write your answer with explanation. [1 mark]

(c) A recurrence relation is an equation that defines a sequence based on a rule the gives the next term as a function of the previous term(s). For example, the factorial function can be defined by this recurrence relation:

$$0! = 1 \qquad \text{and} \qquad n! = n \times (n-1)! \text{ for } n > 0.$$

In an orientation game with girls and boys, chairs are placed in a row and pupils are to sit on them, one person on each chair. The rule, however, states that no two boys are allowed to sit next to each other.

Let there be $n \geq 1$ chairs, and let $f(n)$ be the number of ways the chairs can be filled. For example, let G denote a girl, and B a boy, then for $n = 2$, the chairs may be filled with GG, GB or BG, a total of 3 ways, hence $f(2) = 3$. (i) Write a recurrence relation for $f(n)$, including its initial condition(s), and (ii) what is $f(12)$? [2 marks]

(You do not need to explain how you obtain (i), and do not need to show the working for (ii).)

(d) A bowl contains three coins, two of which are normal coins and one is a two-headed coin. You pick one coin at random and toss it. What is the probability that you get a head? Show your working and write your answer as a fraction. [2 marks]

(e) A bowl contains three coins, two of which are normal coins and one is a two-headed coin. You pick one coin at random, toss it and get a head. What is the probability that the coin is two-headed? Show your working and write your answer as a fraction. [1 mark]

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